

Program : Diploma in Instrumentation Engineering	
Course Code : 6081B	Course Title: Power Plant Instruments
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week : 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To familiarize with the basics of Power sources, Power plant and power Generation.
- To impart the knowledge about the measurements, Analyzers and control loops used in power plant.

Course Prerequisites:

Topic	Course code	Course name	Semester
Measurement Techniques		Sensors and Transducers	2
Process parameters		Process Variable Measurements	4
Control loops		Process Control Instrumentation	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the working of steam power plants and its features.	14	Understanding
CO2	Explain the measurements and control strategies in Feed water Circuit	15	Understanding
CO3	Illustrate the measurements and control strategies in Air – fuel Circuit	15	Understanding
CO4	Describe the measurements and control strategies in Turbine	14	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the working of steam power plants and its features.		
M1.01	Explain different energy sources	2	Understanding
M1.02	Explain Power generation	3	Understanding
M1.03	Explain Nuclear reactor function	4	Understanding
M1.04	Explain Thermal power plant components and operation	5	Understanding
Contents: List and compare various Non conventional energy Sources - steam power plant - layout of typical thermal power plant - generation of electricity in thermal power plant - block diagram of a nuclear power plant - main components of nuclear reactor - define fusion and fission- advantages and disadvantages of nuclear power plant - efficiency of thermal power plant - main components of thermal power plant - define the efficiencies of boiler & turbine alternator - elements and equipments of thermal power plant - flow circuits of thermal power plant.			
CO2	Explain the measurements and control strategies in feed water circuit		
M2.01	Explain boiler feed water measurements and control	2	Understanding
M2.02	Explain Pressure and temperature measurements in power plant	3	Understanding
M2.03	Describe boiler drum level monitoring and control	4	Understanding
M2.04	Illustrate boiler feed water treatment	4	Understanding
M2.05	List various safety interlocks in boiler	2	Understanding
	Series Test – I	1	

Contents:

Boiler feed water measurements and control – pressure – temperature - flow – boiler drum level measurement and control –drum level indicator local and remote - two element control – three element control – five element control – impurities and its effects in feed water – feed water treatment - external and internal treatment - safety interlocks for Boiler operation

CO3	Explain the measurements and control strategies in air- fuel circuit		
M3.01	List the fuel types used in boiler	2	Understanding
M3.02	Explain various parameter measurement in air fuel circuit	3	Understanding
M3.03	Explain various control schemes in air fuel circuit	6	Understanding
M3.04	Explain the Optimization of combustion control	2	Understanding
M3.05	Illustrate the working of analyzers, dust and smoke detectors.	2	Understanding

Contents:

Types of fuels used in boiler - constituents of flue gases & waste gases - measurements in air fuel circuit - the combustion control with gas fuel air - fuel ratio control - combustion control with liquid fuel - air fuel ratio control - optimization of combustion control - furnace draught control -forced draught control - induced draught control - Balanced draught control - Zirconia oxygen analyzer - the Combustibles (Co+H₂) analyzer - Smoke detector - Dust monitor.

CO4	Illustrate the measurements and control strategies in Turbine		
M4.01	Explain the working and classification of Turbines	3	Understanding
M4.02	Describe various measurements in Turbines	3	Understanding
M4.03	Explain the control systems turbine	3	Understanding
M4.04	Describe the turbine lubrication system and control	3	Understanding
M4.05	Explain the master control for power station	2	Understanding
	Series Test – II	1	

Contents:

Turbine- working - classification of turbine - measurements in Turbine – process – mechanical and electrical parameters – Turbine controls - lubrication system for turbo – alternator- Master control for power station.

Text / Reference:

T/R	Book Title/Author
R1	K.Krishnaswamy and M.Ponnibala- Power Plant Instrumentation- PHI Learning Pvt Ltd, 2011
R2	B.G.Liptak-Instrumentation Engineers Handbook (Process Measurement and Analysis), Fourth Edition- Chilton Book Co, 2003
R3	John G Webster-The Measurement, Instrumentation, and Sensors Handbook- CRC and IEEE Press, 1999.
R4	Sam. G. Dukelow- Control of Boilers- Instrument Society of America

Online Resources:

Sl. No	Website Link
1	https://www.oreilly.com/library/view/power-plant-instrumentation/9780128009406/
2	http://www.powerplantinstrumentationcontrol.yolasite.com/power-plant-details/1-classification-of-instruments-in-power-plant
3	https://nptel.ac.in/courses/108/105/108105064/